

BAAZ JKBOSE



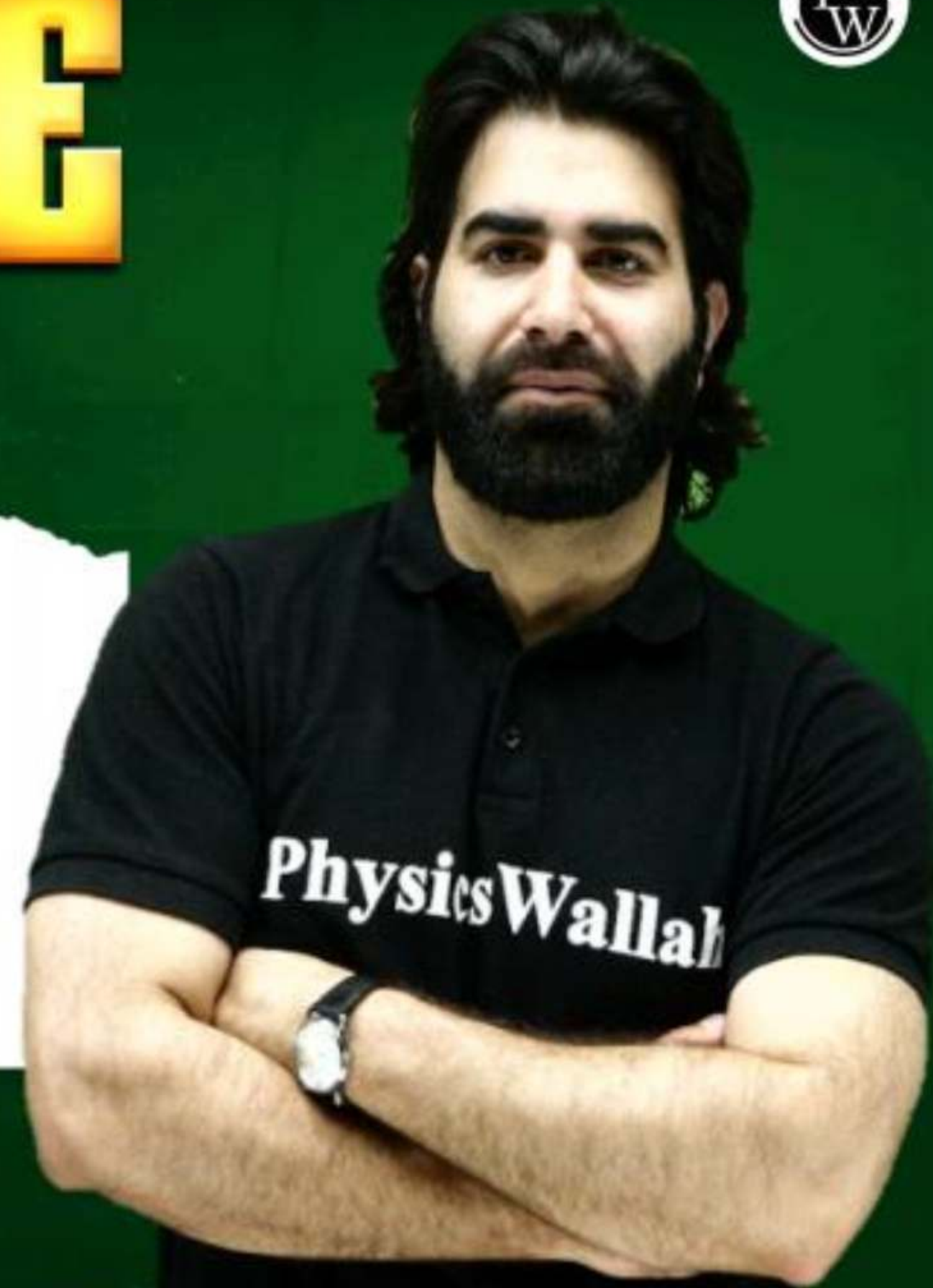
2026 CLASS-11TH

Body Fluids and Circulation

LECTURE-05

Zoology

Dr Adil Altaf sir



Topics To Be Covered

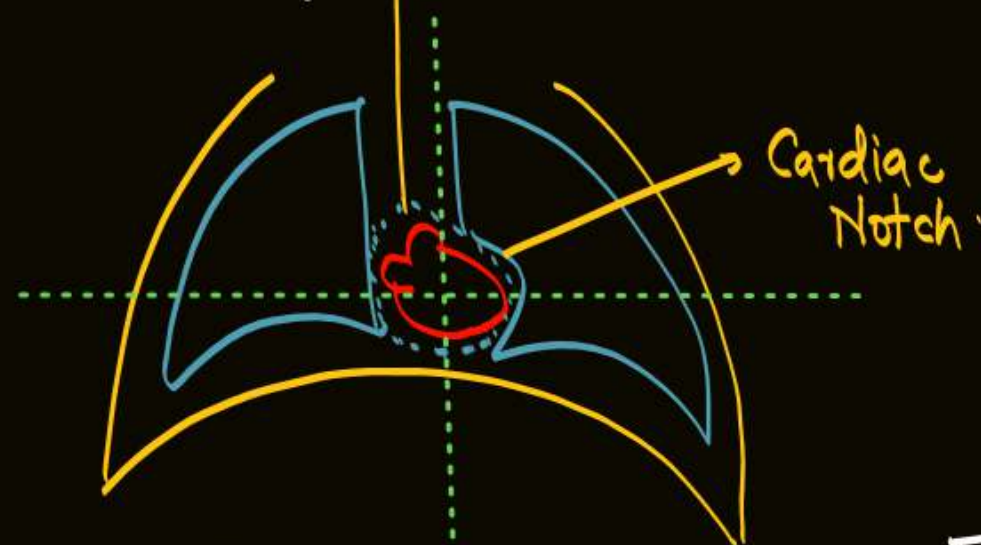
- 1 Human Circulation, Human Heart ,

Human Heart

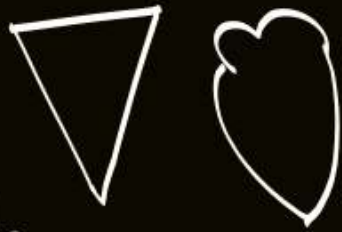
→ 4 chambers.

→ Size = clenched fist.

→ Location = Mediastinum
of Thoracic cavity.



- Heart is Roughly pyramidal in shape.
- It Apex is tilted towards Left Side.
- Heart is Mesodermal in origin.

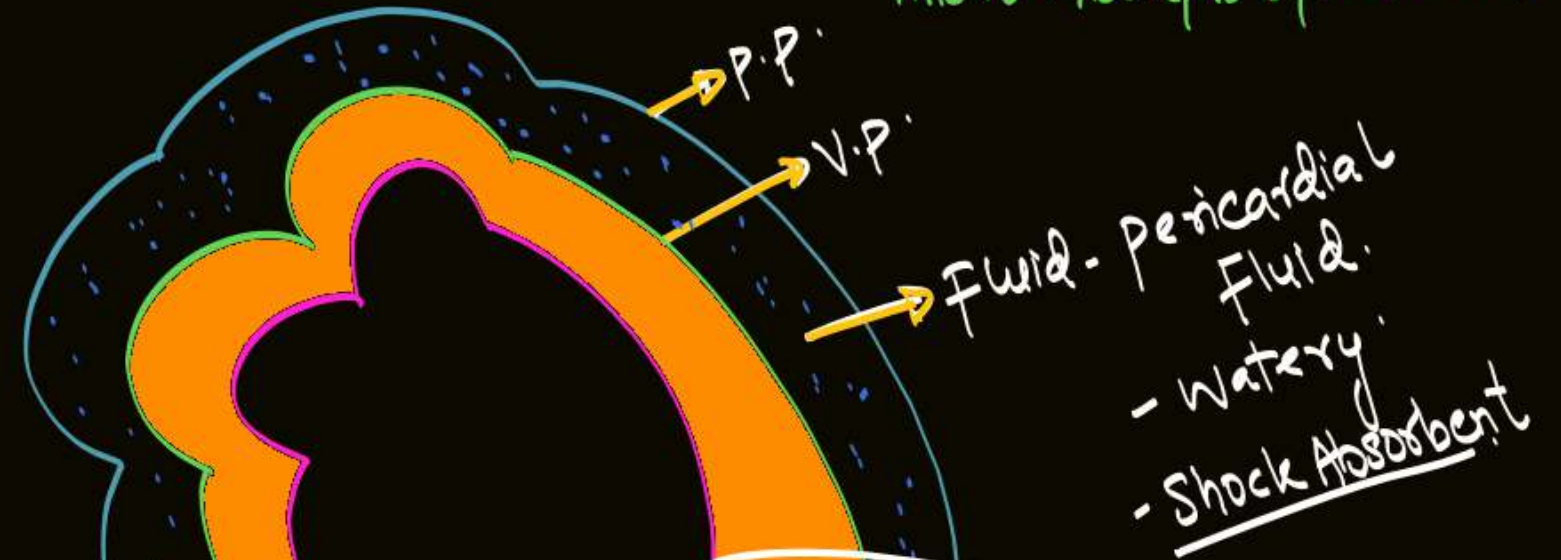


→ Covering of Heart.

2-Covering →

- outer → Parietal pericardium.
- Space → pericardial space.
- Inner → Viseral pericardium.

This is Also k/as Epicardium.



→ Wall of Heart - 3 layers -

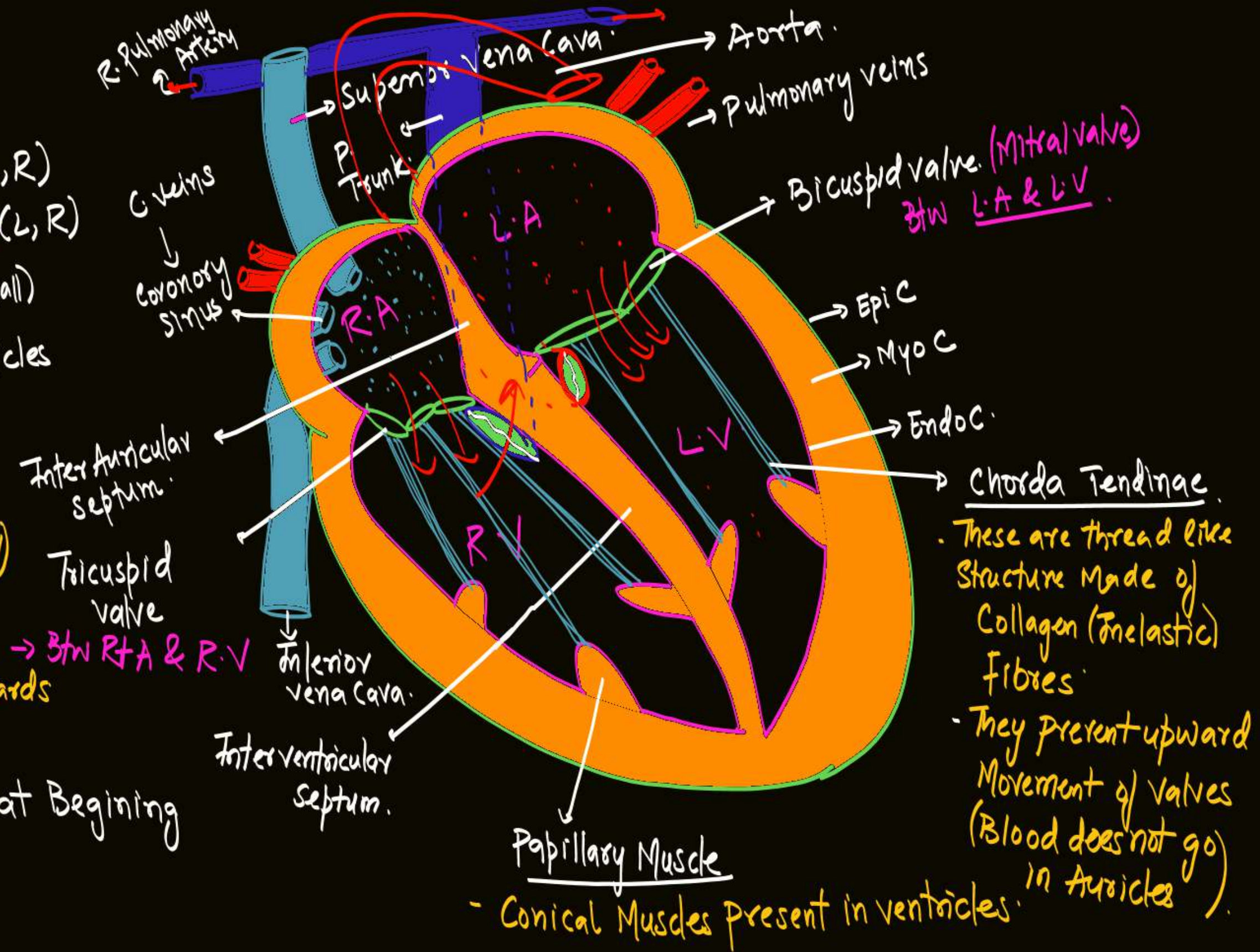
- ① Epicardium.
- ② Myocardium.
- ③ Endocardium.

- Epicardium - outer layer of Heart
 - Made of Simple Squamous Epithelium.
- Myocardium - Middle and Muscular layer of Heart
 - Made of Cardiac Muscle cells.
 - ↳ Autoexcitability (can contract by their own).
(By generating own signal.)
 - ↳ Makes Human Heart - Myogenic
(Muscle generate own signal.)
- Endocardium → Inner layer
 - Made of Simple Squamous epithelium.

Heart of insects
Neurogenic (cockroach).
Neuron generate signal.

Structure of Heart

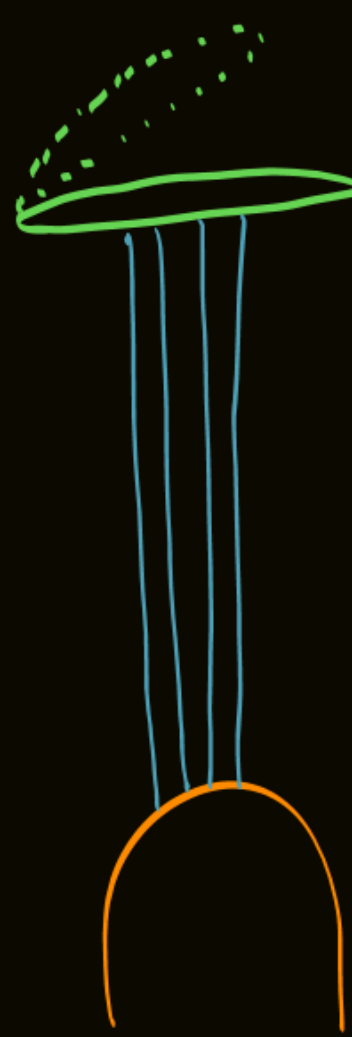
- Four chambers — 2 Atrium (L, R)
— 2 ventricles (L, R)
- chambers separated by septum (wall)
- wall of ventricles Thicker than Atricles
- Thickest wall (left ventricle).
- Atricles & ventricles are separated by Atrioventricular (A.V) Valves (Bicuspid & tricuspid).
- These valves open downwards Towards ventricles.
- Semilunar Valves present at Beginning of P. Artery and Aorta.



- open valve



closed valve



Conduction System of Heart

↳ Heart generates its own Signal.
↳ It is due to presence of Special Musculature
k/as - Nodal Tissue

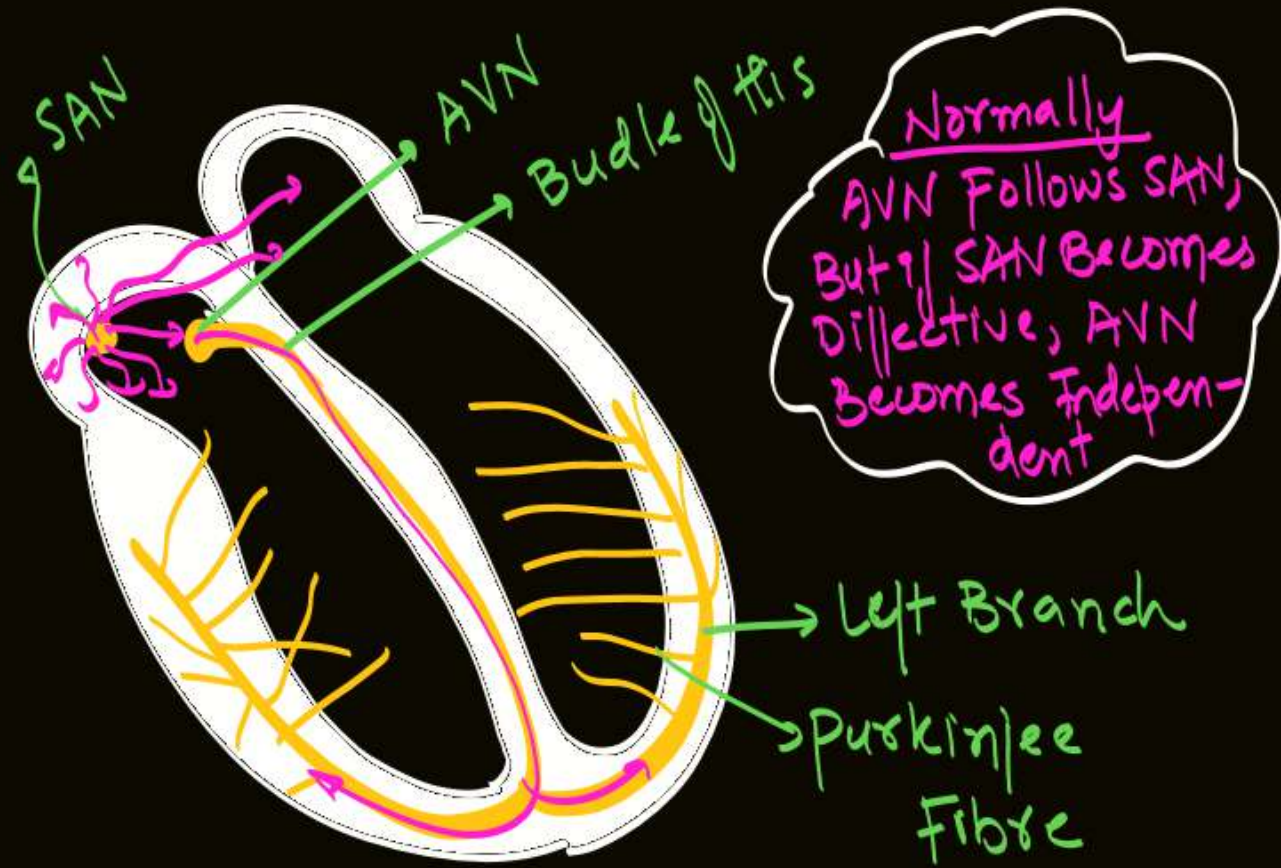
↳ They Include - SA Node (Sinoatrial Node)
- AV Node (Atrioventricular Node).

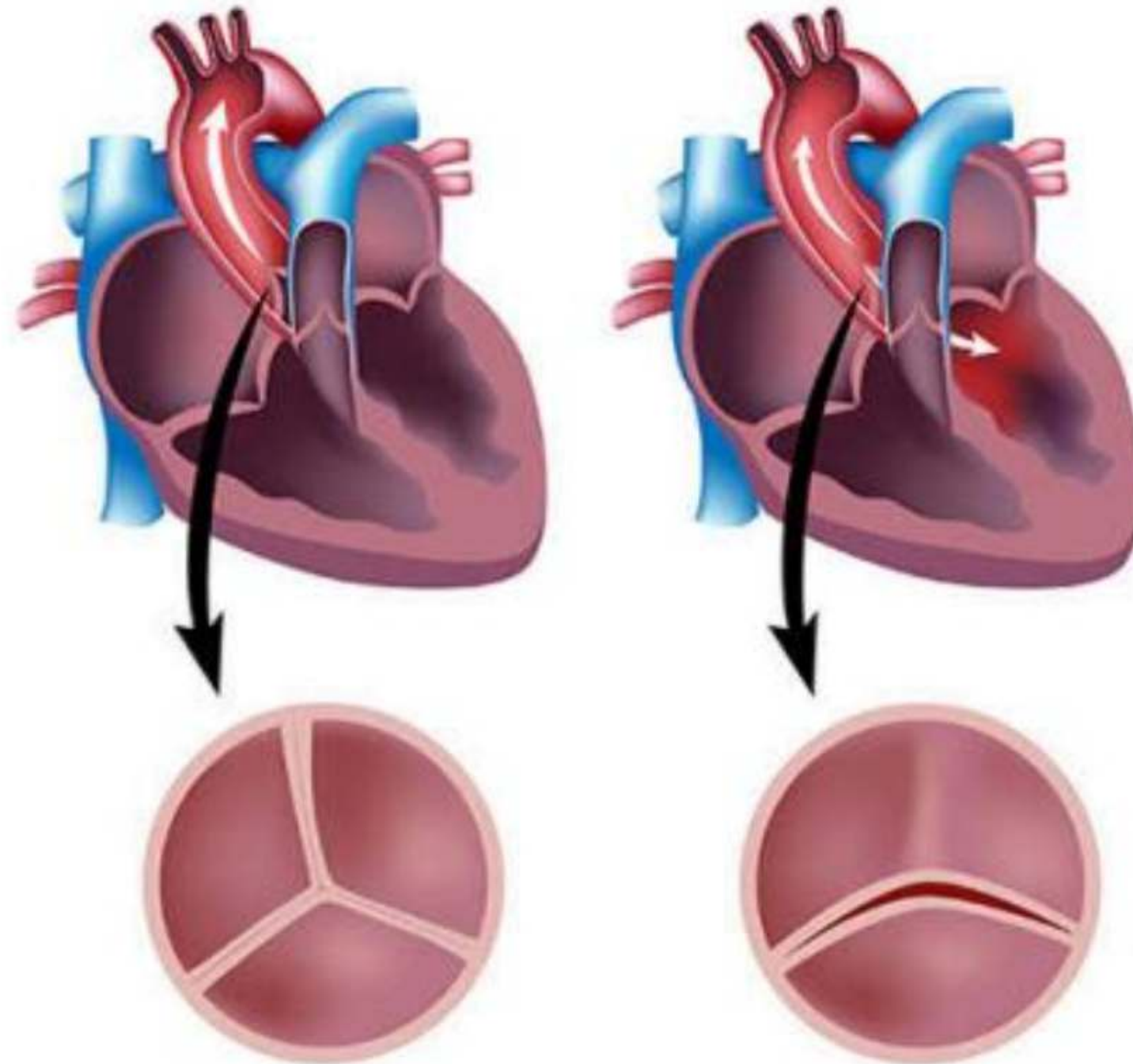
→ SAN

- located at upper Rt corner of Rt Atrium.
- Also known as Pace Maker.
- Generates Impulse (Signal) which spreads in Atria.
- Causes contraction of Atrium.

→ AVN (Pace keeper)

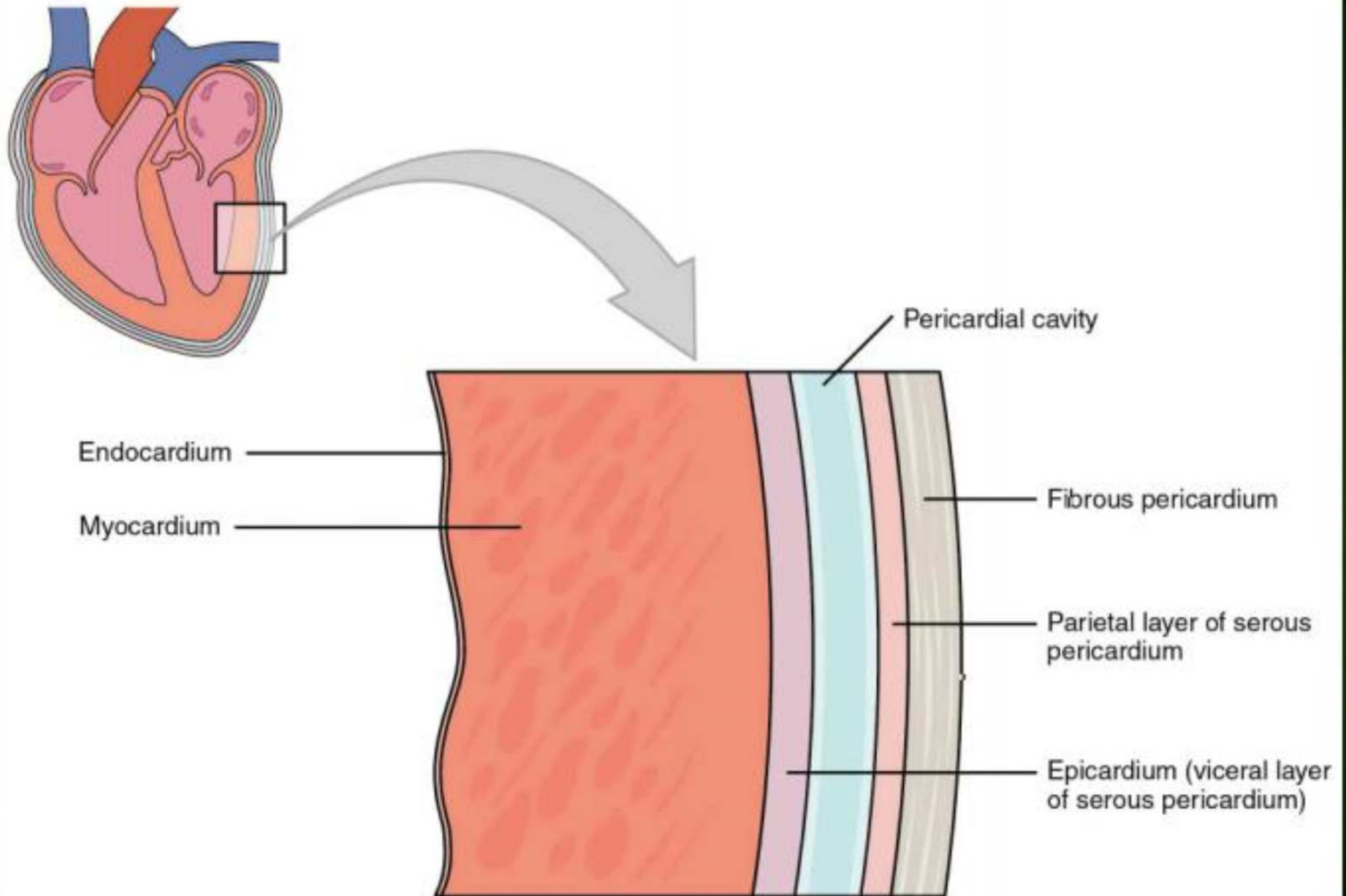
- located at lower left corner of Rt Atrium.
- It continues as Bundle of His.
- Which further divides into Rt & left Bundle Branch and finally Purkinjee Fibre.
- Signal generated spreads in ventricles.

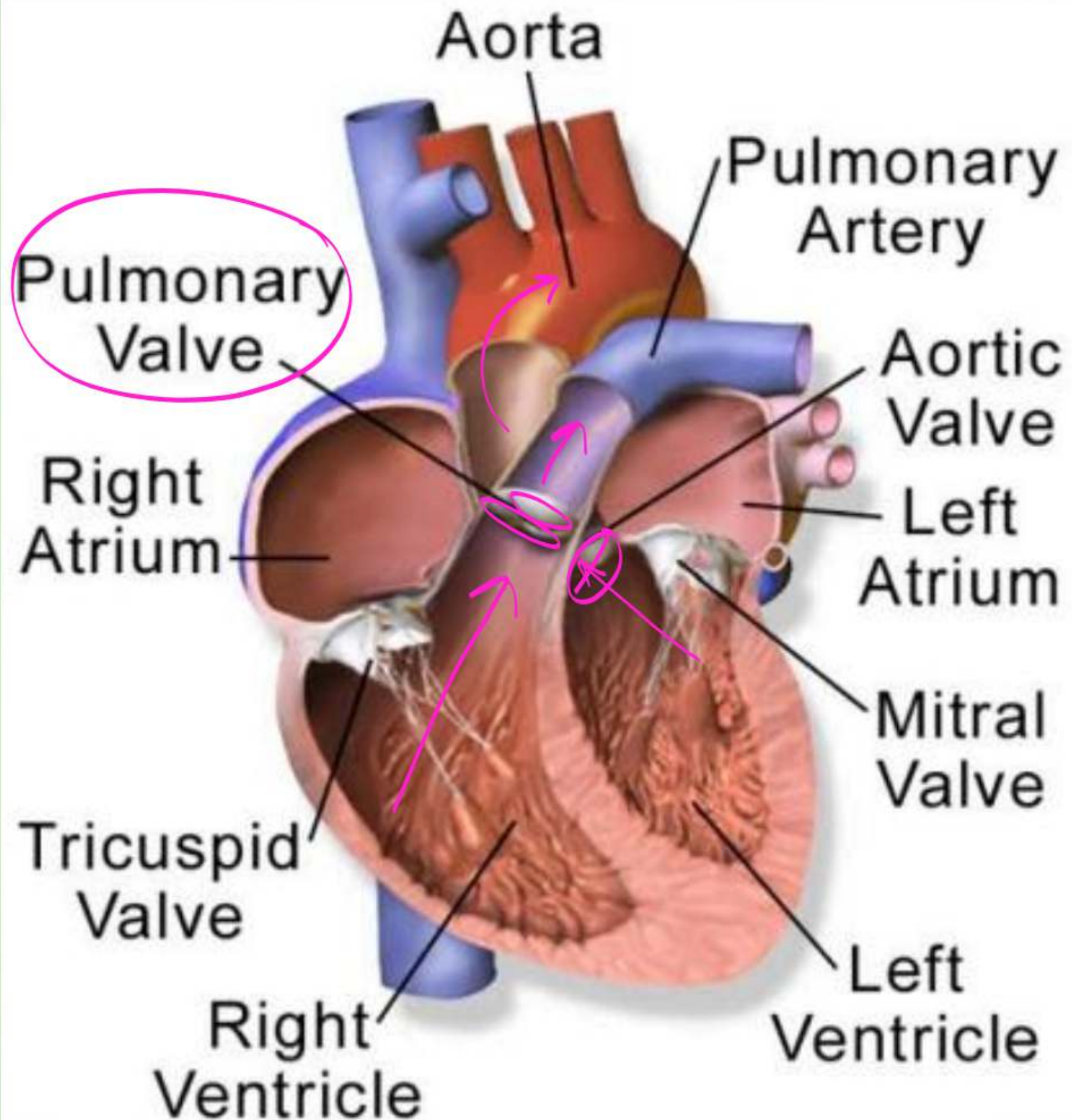


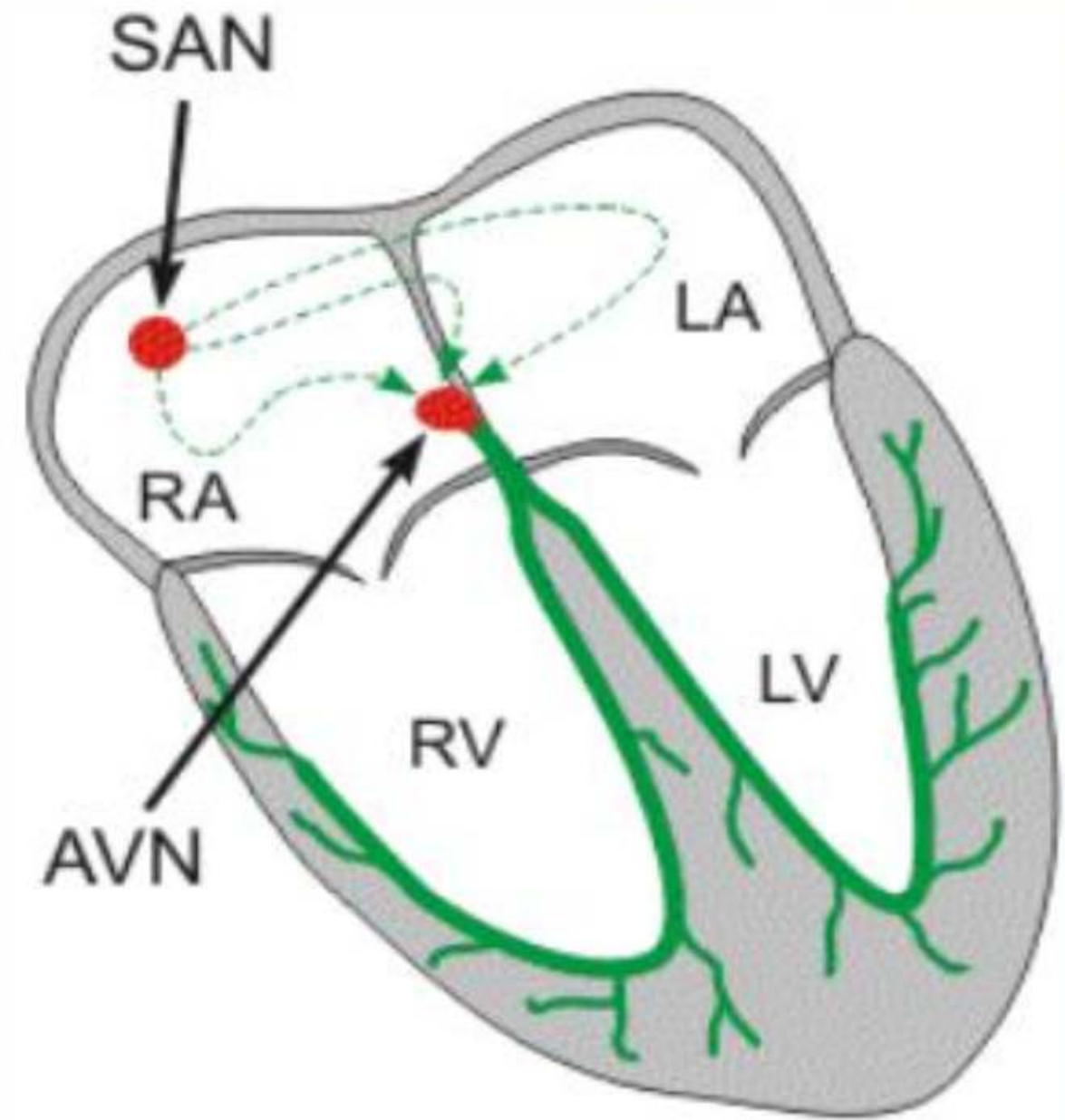


Normal tricuspid valve

Bicuspid aortic valve



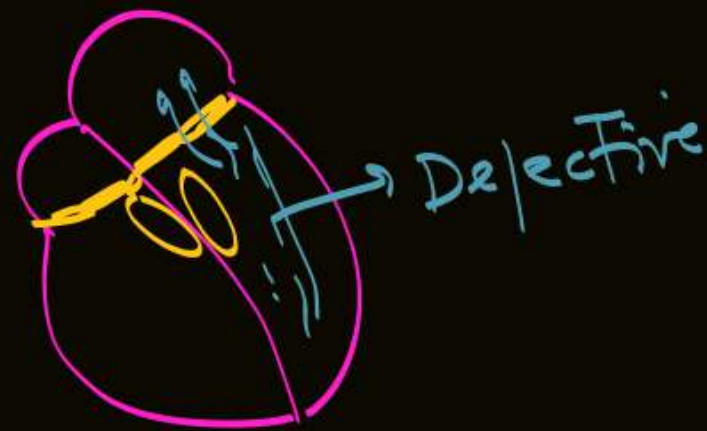




SAN, sinoatrial node; AVN, atrio-ventricular node; RA, right atrium; LA, left atrium, RV, right ventricle; LV, left ventricle.

Qno1. If the chordae Tendinae of Left ventricle are defective, what is most likely going to happen;

- (a) Less Blood in P. Artery. X
- (b) Blood Flows Back to Right Atricle. X
- (c) No change in Blood pumping into Aorta X
- (d) Less Blood pumping in Aorta. ✓



Qno2 - Which is 'incorrectly' Matched.

- (a) Mitral Valve = Btw L.A & L.V. ✓
- (b) Bicuspid Valve = Btw R.A & R.V. ✗
- (c) Semilunar valve = Pulmonary Artery ✓
- (d) Tricuspid valve = Btw R.A & R.V. ✓

Thank
You